

SnowPro Specialty: Gen AI — Prep Guide

PJ's Academy · Prove you can build GenAI apps on Snowflake Cortex.

The Gen AI Specialty exam (GES-C01) validates skills in building LLM-powered applications on Snowflake using **Cortex** — LLM functions, RAG with Cortex Search, agents, and Snowflake Intelligence.

Prerequisite: SnowPro Core certified.

Exam At A Glance

Item	Detail
Exam code	GES-C01
Questions	~55
Duration	85 minutes
Cost	\$225 USD
Prerequisite	SnowPro Core

Domain Weightings

#	Domain	Weight
1	Snowflake Gen AI & LLM Fundamentals	25%
2	Cortex LLM Functions & Prompting	30%
3	RAG, Cortex Search & Agents	30%
4	Governance, Cost & Evaluation	15%

Domain 1 — Gen AI Fundamentals (25%)

- LLM basics: tokens, context window, temperature, embeddings, hallucination.
- Snowflake Cortex overview — serverless, governed, in-platform GenAI.
- Available models (Cortex hosts multiple LLMs — Llama, Mistral, Snowflake Arctic, Claude, etc.).
- When to use prompting vs RAG vs fine-tuning.

Domain 2 — Cortex LLM Functions & Prompting (30%)

Core functions

```
SELECT SNOWFLAKE.CORTEX.COMPLETE('llama3.1-70b', 'Explain micro-partitions simply');
SELECT SNOWFLAKE.CORTEX.SENTIMENT(review) FROM reviews;
SELECT SNOWFLAKE.CORTEX.SUMMARIZE(article) FROM docs;
SELECT SNOWFLAKE.CORTEX.TRANSLATE(text, 'en', 'hi') FROM messages;
SELECT SNOWFLAKE.CORTEX.EXTRACT_ANSWER(context, 'What is the refund policy?');
SELECT SNOWFLAKE.CORTEX.EMBED_TEXT_768('e5-base-v2', content) FROM docs;
```

Prompting

- Structured prompts, few-shot examples, JSON-mode / structured outputs.
- **COMPLETE** with options (temperature, max_tokens, response_format, guardrails).
- Cortex **AISQL** — **AI_COMPLETE** , **AI_CLASSIFY** , **AI_FILTER** , **AI_AGG** , **AI_SUMMARIZE_AGG** .

Domain 3 — RAG, Cortex Search & Agents (30%)

Cortex Search

- Fully-managed hybrid (vector + keyword) search service for RAG.
- Create a search service over a table; query for relevant chunks.

```
CREATE CORTEX SEARCH SERVICE docs_search
ON content ATTRIBUTES title
WAREHOUSE = search_wh TARGET_LAG = '1 hour'
AS SELECT content, title FROM knowledge_base;
```

RAG pattern

1. Chunk & embed documents (or use Cortex Search which manages this).
2. Retrieve relevant chunks for a question.
3. **COMPLETE** with retrieved context → grounded answer.

Cortex Analyst & Agents

- **Cortex Analyst** — text-to-SQL over a semantic model (natural-language BI).
- **Cortex Agents / Snowflake Intelligence** — orchestrate tools (Search + Analyst) to answer complex questions.

Domain 4 — Governance, Cost & Evaluation (15%)

- Data stays in Snowflake — RBAC/masking still apply to GenAI.
- Cost model — Cortex functions billed by tokens/credits.
- **Cortex Guard** — safety filtering.
- Evaluation — relevance, groundedness, hallucination checks; **AI_*** eval helpers.

High-Yield Facts

- **Cortex Search** = managed hybrid retrieval for RAG (no external vector DB needed).
- **Cortex Analyst** = text-to-SQL over a semantic model.
- **COMPLETE** is the general LLM call; task functions (SENTIMENT, SUMMARIZE) are shortcuts.
- **EMBED_TEXT_768** / **EMBED_TEXT_1024** generate embeddings for similarity.
- **Cortex Guard** filters unsafe content.

- Data never leaves Snowflake — governance (RBAC, masking) still applies to GenAI.
 - **AI SQL** (`AI_CLASSIFY` , `AI_FILTER` , `AI_AGG`) brings LLMs into set-based SQL.
-

Practice Questions (10)

- Q1.** The general-purpose Cortex LLM call is: A) SUMMARIZE B) COMPLETE C) SENTIMENT D) TRANSLATE
- Q2.** Managed hybrid retrieval for RAG is provided by: A) Cortex Search B) Snowpipe C) Streams D) Marketplace
- Q3.** Natural-language-to-SQL over a semantic model is: A) Cortex Search B) Cortex Analyst C) Cortex Guard D) COMPLETE
- Q4.** Which generates text embeddings? A) COMPLETE B) `EMBED_TEXT_768` C) SENTIMENT D) `EXTRACT_ANSWER`
- Q5.** RAG grounds an LLM answer by: A) Fine-tuning B) Retrieving relevant context then completing C) Higher temperature D) More tokens
- Q6.** Unsafe-content filtering in Cortex is: A) Cortex Guard B) Masking policy C) Row policy D) Network policy
- Q7.** Cortex functions are billed primarily by: A) Rows B) Tokens/credits C) Storage D) Warehouses
- Q8.** Data used by Cortex: A) Leaves Snowflake B) Stays in Snowflake (governed) C) Goes to OpenAI D) Is public
- Q9.** Which brings LLM classification into set-based SQL? A) `AI_CLASSIFY` B) `COPY` C) `FLATTEN` D) `PIVOT`
- Q10.** Orchestrating Search + Analyst as tools describes: A) Cortex Agents B) Snowpipe C) Streams D) Clone

Answers

1-B · 2-A · 3-B · 4-B · 5-B · 6-A · 7-B · 8-B · 9-A · 10-A

2-Week Plan

- **Week 1:** LLM fundamentals + Cortex functions + prompting.
 - **Week 2:** RAG (Cortex Search), Analyst/Agents, governance + practice.
-

 *Snowflake Mastery — PJ's Academy*
